Skills Summary

Postulates, Theorems, Converses and Biconditionals

Use this reference while completing your worksheet or when studying.

The four labels. **Definition** — states what a term means; reversible by agreement. **Postulate** — assumed without proof, so the system has a starting point. **Theorem** — proved from definitions, postulates and earlier theorems. **Biconditional** — "p if and only if q", valid only when $p \to q$ and $q \to p$ are *both* true.

The converse of "if p, then q" is "if q, then p". Swapping is free; *truth is not carried along.* One counterexample kills a converse.

1. Applying a postulate to compute

Two postulates do most of the algebra in this unit.
Segment Addition: if B is between A and C, then $AB + BC = AC$.
Angle Addition: if D is interior to $\angle ABC$, then $m\angle ABD + m\angle DBC = m\angle ABC$.
Name the postulate first, then write the equation it licenses.

Example: B is between A and C, $AB = x + 4$, $BC = 2x$, $AC = 22$. Find x.
Step 1. Betweenness is what licenses adding the parts, so the Segment Addition Postulate supplies the equation before any algebra happens.
Step 2. $(x + 4) + 2x = 22$ gives $3x + 4 = 22$, so $3x = 18$.

$$\boldsymbol{x = 6} \quad \text{(check: } 10 + 12 = 22\text{)}$$

Try it: $\overrightarrow{BD}$ is interior to $\angle ABC$ with $m\angle ABD = 3x$, $m\angle DBC = 2x + 10$, $m\angle ABC = 85°$. Find x.

check: $\boldsymbol{x = 15}$

2. Postulate, theorem, or definition?

Ask two questions.
Could it be proved from something earlier? No → postulate. Yes → theorem.
Is it telling me what a word means, both ways? → definition.
Obviousness decides nothing: "vertical angles are congruent" is obvious and is still a theorem, because it has a proof.

Example: Classify *through any three non-collinear points there is exactly one plane.*
Step 1. Start with the provability question, since that is the only thing that separates a postulate from a theorem.
Step 2. There is no earlier statement about planes to derive it from — it is one of the assumptions that gives the word *plane* its behaviour, and it is not unpacking a definition. So it is a **postulate**. *(Classification is read by a person, not computed — flagged for manual review.)*

Try it: Classify *a segment bisector is a segment, ray or line that divides a segment into two congruent segments*, and give the reason.

check: definition — it states what the term means, and it reverses

3. Testing the truth of a converse

To test "if q, then p", hunt for one case where q holds and p fails. With an equation, that means solving **completely**: if the solution set is bigger than the value named, the extra solution *is* the counterexample.

Example: Is the converse of "if $x = 7$, then $x^2 = 49$" true?
Step 1. The converse claims $x^2 = 49$ forces $x = 7$, so the test is whether 7 is the only number whose square is 49 — solve, do not substitute.
Step 2. $x^2 - 49 = 0$ factors as $(x - 7)(x + 7) = 0$, giving two solutions.

$$\boldsymbol{x = -7} \text{ or } \boldsymbol{x = 7}$$

$x = -7$ satisfies the hypothesis and not the conclusion, so the converse is **false**.

Try it: Is the converse of "if $x = 5$, then $3x = 15$" true? Solve $3x = 15$ completely and decide.

check: $\boldsymbol{x = 5}$ only, so the converse is true

4. When a biconditional is valid

"p if and only if q" is valid exactly when the two sides describe the **same set of cases**. Check the conditional and its converse separately; if either fails, the biconditional fails. Definitions always pass this test — that is what makes them definitions.

Example: Is "$5x + 2 = 17$ if and only if $x = 3$" valid?
Step 1. Two sides, two directions — compare the solution sets rather than checking one substitution.
Step 2. $5x + 2 = 17$ gives $5x = 15$, whose solution set is a single value; and substituting that value returns 17, so the reverse direction holds too.

$$\boldsymbol{x = 3} \quad \text{— both directions hold, so the biconditional is valid.}$$

Try it: Is "$x^2 - 2x - 8 = 0$ if and only if $x = 4$" valid? Solve completely first.

check: $\boldsymbol{x = -2}$ or $\boldsymbol{x = 4}$, so it is not valid